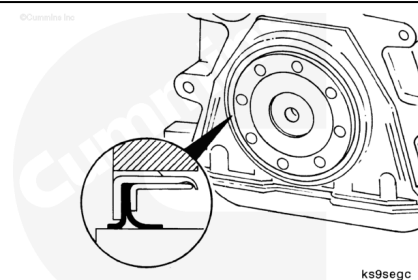


Trainings ▾

General Information

A lip style rear crankshaft seal (illustrated) is used, in which the rotating portion of the sealing occurs at the contact surface between the lip of the seal and the crankshaft.

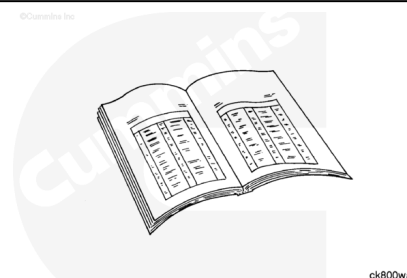
For front gear train engines, the rear crankshaft seal is mounted in a rear seal carrier that bolts to the rear of the cylinder block (illustrated). For rear gear train engines, the rear crankshaft seal is installed in the flywheel housing bore (**not** illustrated).



Preparatory Steps

⚠ WARNING ⚠

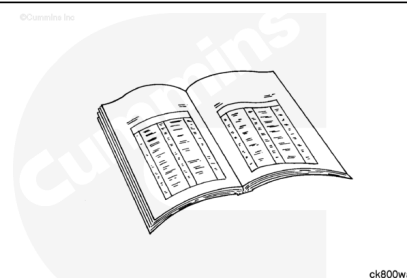
Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



- Disconnect the batteries. Refer to the OEM service manual.

Note : Use a container that can hold at least 26 liters [27 qt] of lubricating oil.

- If equipped with a wet flywheel housing, drain the oil from the flywheel housing by removing the plug in the bottom of the flywheel housing
- For front gear train engines, remove the starting motor. Refer to Procedure 013-020 in Section 13.
(/qs3/pubsys2/xml/en/procedures/40/40-013-020-tr.html)



- Remove the transmission and all related components (if equipped). Refer to the OEM instructions
- Remove the flywheel/flexplate assembly. Refer to Procedure 016-005 in Section 16 (</qs3/pubsys2/xml/en/procedures/40/40-016-005-tr.html>) or Refer to Procedure 016-004 in Section 16 (</qs3/pubsys2/xml/en/procedures/40/40-016-004-tr.html>)
- For rear gear train engines, remove the rear crankshaft seal. Refer to Procedure 001-024 in Section 1 (</qs3/pubsys2/xml/en/procedures/40/40-001-024-tr.html>)
- For some engines, it may be necessary to remove the crankcase breather tube. Refer to Procedure 003-018 in Section 3 (</qs3/pubsys2/xml/en/procedures/40/40-003-018-tr.html>)
- Remove any OEM attached components (mufflers, shift mechanisms, air filters, etc.) to the flywheel housing. Refer to the OEM instructions
- For front gear train engines, remove the flywheel housing. Refer to Procedure 016-006 in Section 16 (</qs3/pubsys2/xml/en/procedures/40/40-016-006-tr.html>).

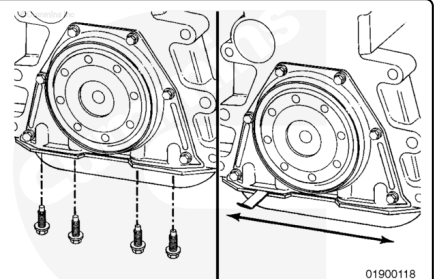
Remove

Front Gear Train

Remove the four oil pan mounting capscrews that secure the oil pan to the rear cover.

⚠ CAUTION ⚠

Use extreme care when releasing the oil pan gasket from the rear cover to prevent damage to the gasket.

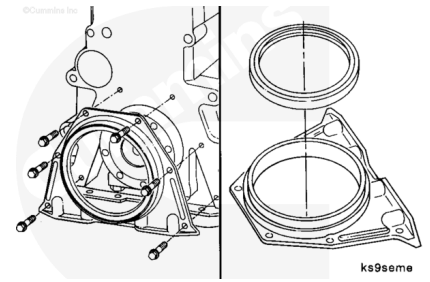


Insert the feeler gauge or shim stock between the rear cover and the oil pan gasket. Move the feeler gauge back and forth to release the gasket from the rear cover.

Remove the capscrews from the rear cover.

Remove the cover from the crankshaft flange.

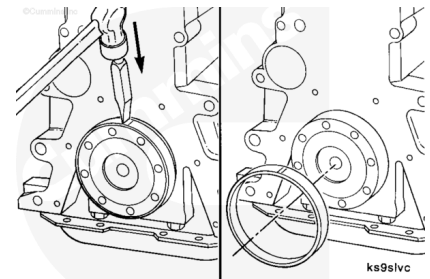
Support the rear seal carrier on a flat work surface with wooden blocks. Use a suitable punch and hammer to drive the old seal out of the rear seal carrier.



All Applications

⚠ CAUTION ⚠

Do not nick or gouge the crankshaft with the chisel. If the crankshaft is damaged, it must be replaced.



Note : For rear gear train engines, if a wear sleeve has previously been installed, the flywheel housing **must** be removed to remove the wear sleeve. After removing the wear sleeve, reinstall the flywheel housing. Refer to **Refer to Procedure 016-006 in Section 16** (</qs3/pubsys2/xml/en/procedures/40/40-016-006-tr.html>).

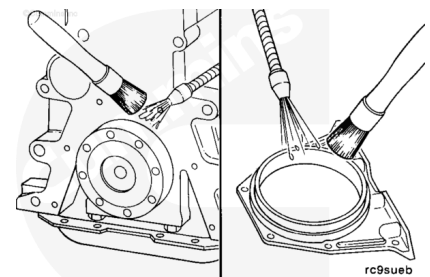
If a wear sleeve has previously been installed, use a dull chisel that is **only** as wide as the wear sleeve.

Make one or two soft blows with a hammer to make chisel marks across the wear sleeve. This will expand the wear sleeve, allowing the sleeve to be removed.

Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to avoid personal injury.



⚠ WARNING ⚠

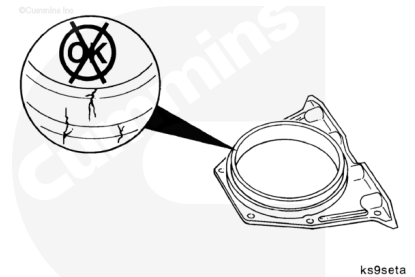
Compressed air used for cleaning should not exceed 207 kPa [30 psi]. Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause bodily

injury.

For front gear train engines, clean the gasket surface of the cylinder block and rear seal carrier.

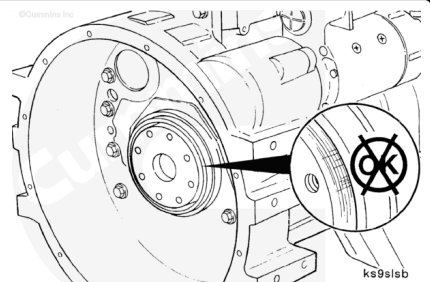
Dry these areas with compressed air.

If equipped, inspect the rear seal carrier for cracks or other damage. Replace the rear seal carrier if any damage is found.



For rear gear train engines, clean the flywheel housing bore of any seal residue.

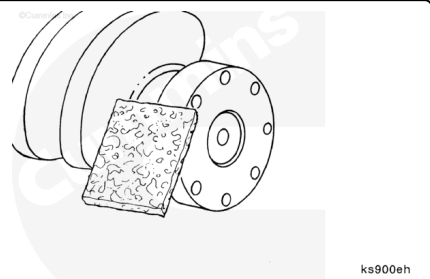
Inspect the flywheel housing bore for nicks or burrs. Use an abrasive pad, Part Number 3823258 or equivalent, to remove any minor damage.



Use a fine crocus cloth to remove any rust or other deposits from the crankshaft flange.

Use a clean cloth and solvent to clean the crankshaft flange.

Inspect the crankshaft flange for nicks or burrs. Use an abrasive pad, Part Number 3823258 or equivalent, to remove any minor damage.

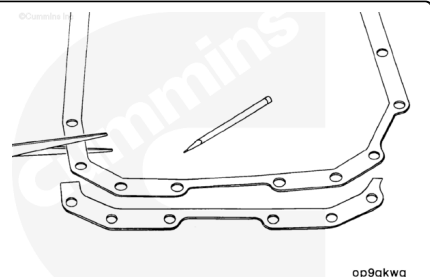


Install

Front Gear Train

Service Tip: If the oil pan gasket was damaged during removal of the rear seal carrier, remove the damaged gasket and:

1. If previously equipped with a paper oil pan gasket, use the old gasket as a pattern and cut a section of a new gasket to the same size. Use a light coat of sealant, Part Number 3164067, to hold the gasket in place



2. If previously equipped with a sealant only oil pan gasket, apply a bead of sealant, Part Number 3164070, to the oil pan flange
3. If previously equipped with a suspended oil pan, the oil pan must be removed and a new gasket installed. Refer to Procedure 007-025 in Section 7 (</qs3/pubsys2/xml/en/procedures/40/40-007-025-tr.html>).

Note : The rear seal carrier must be installed within 10 minutes of applying the sealant.

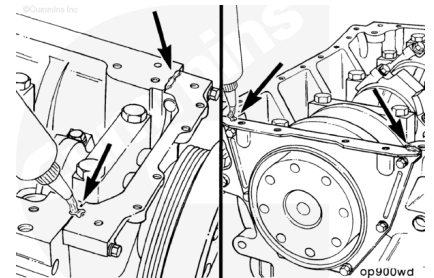
Note : If the oil pan is installed, it may be necessary to loosen additional oil pan capscrews to allow for rear seal carrier and gasket clearance.

Note : It may be necessary to trim the rear seal carrier gasket so that it is even with the oil pan mounting surface. Also, apply a thin bead of sealant, Part Number 3164067, at this joint prior to installing the rear seal carrier. The rear seal carrier must be installed within 10 minutes of applying the sealant.

Install the rear seal carrier, mounting capscrews, and gasket.

Apply Loctite™ 262 or 271 to the mounting capscrews and loosely tighten the rear seal carrier to the block.

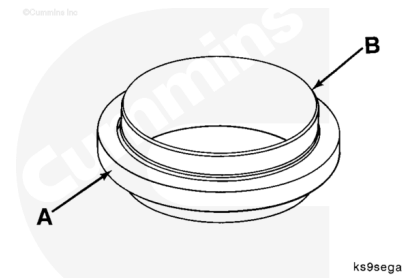
Note : The seal installation will properly align the rear seal carrier.



All Applications

Note : For engines equipped with a wet flywheel housing, make sure to replace the rear crankshaft seal with the correct rear seal. Rear crankshaft seals for a wet and dry flywheel housings may not be the same.

Note : The replacement oversize seal may appear different than the standard size seal.



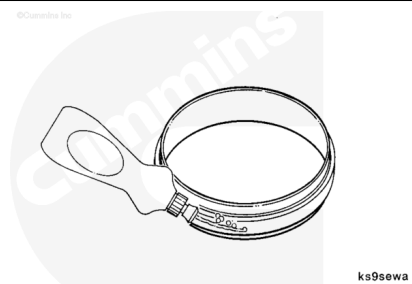
⚠ CAUTION ⚠

The new rear crankshaft seal should not be removed from the crankshaft rear seal wear sleeve. Damage to the sealing lips may occur when reinstalling the seal onto the wear sleeve.

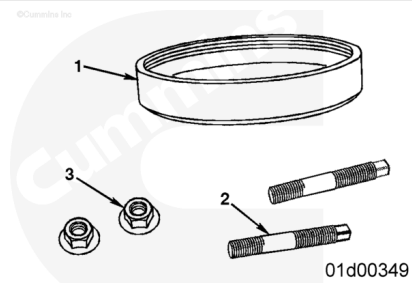
The new oversize seal and wear sleeve comes pre-assembled and will be installed on the crankshaft as an assembly.

- A. Crankshaft oil seal
- B. Wear sleeve.

To aid in installation, the lubricating oil seal requires the application of a mild soap on the outside diameter of the seal case.



Use service tool, Part Number 3824078, to install the crankshaft seal/wear sleeve assembly.



Reference Number	Part Number	Description
1	3163734	Plate
2	3163628	Stud
3	3163741	Hex Nut

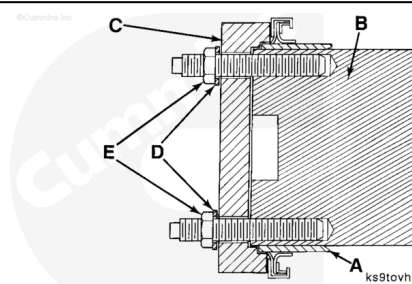
Install two (2) threaded studs into the crankshaft capscrew holes.

Apply a small amount of clean 15W-40 engine oil to the crankshaft, threaded studs, and the inside of the crankshaft rear seal/wear sleeve installation tool.

Position the chamfered end of the wear sleeve (A) onto the end of the crankshaft (B).

Position the counterbore end of installation tool (C) over threaded studs and align with the wear sleeve, perpendicular to the end of the crankshaft.

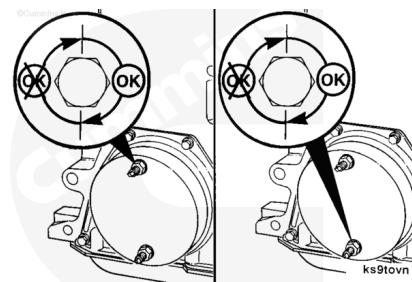
Install the washers (D) and nuts (E) onto the threaded studs.



Note : The illustration shows a front gear train application with a rear crankshaft seal carrier. For rear gear train engines, the procedure for installing the crankshaft wear sleeve/seal into the flywheel housing bore is the same.

Alternately tighten the nuts ½ of a turn until the installation tool contacts the end of the crankshaft.

Do **not** exceed ½ of a turn of each nut to prevent wear sleeve binding and irregular stretch.



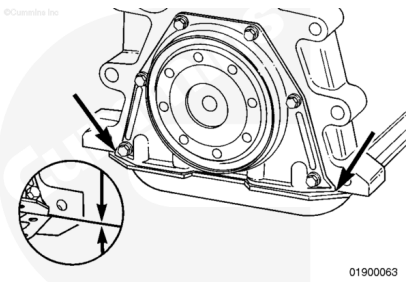
Torque Value: 20 n•m [177 ft-lb]

Remove the installation tool and the threaded studs.

Front Gear Train

⚠ CAUTION ⚠

Do not push or force the cover in any direction. This may cause an irregular seal lip position after seal installation. An engine oil leak will result.

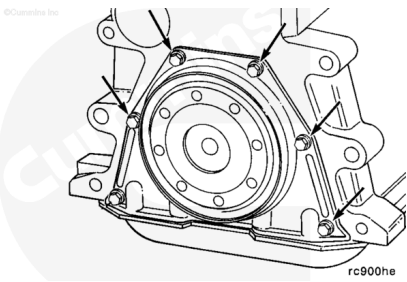


Align the rear cover even with both sides of the oil pan rail on the cylinder block.

Tighten the rear cover capscrews.

The chart below show the proper torque value when using either 9.8 or 10.9 grade capscrew. The grade is embossed on the top of each capscrew.

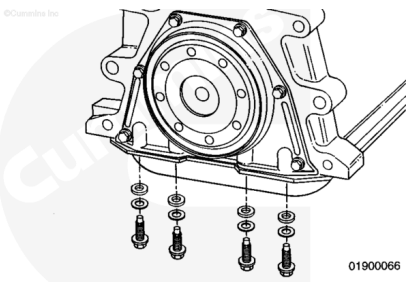
Apply Loctite™ 262 or 271 to both part numbers below during installation.



Rear Seal Carrier				
Bolt Number	Bolt Size	Bolt Class	Torque	Loctite™
3913638	M-6	9.8	10 N•m [89 in-lb]	Yes
3991306	M-6	10.9	13 N•m [115 in-lb]	Yes

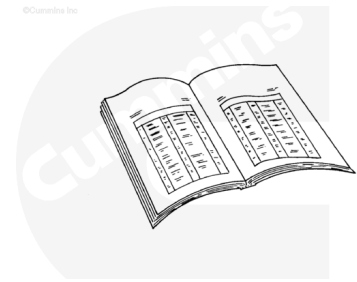
Install the four rear oil pan mounting capscrews into the pan.

Torque Value: 24 n•m [212 in-lb]



Finishing Steps

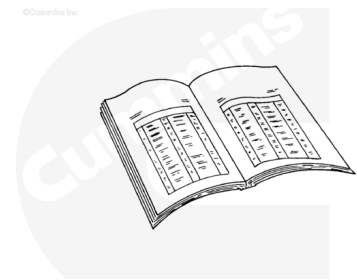
- If previously removed, install the crankcase breather tube. Refer to Procedure 003-018 in Section 18 (</qs3/pubsys2/xml/en/procedures/40/40-003-018-tr.html>)
- Install the flywheel/flexplate assembly. Refer to Procedure 016-005 (</qs3/pubsys2/xml/en/procedures/40/40-016-005-tr.html>) in Section 16 or Refer to Procedure 016-004 in Section 16. (</qs3/pubsys2/xml/en/procedures/40/40-016-004-tr.html>)
- For front gear train engines, install the starting motor. Refer to Procedure 013-020 in Section 13 (</qs3/pubsys2/xml/en/procedures/40/40-013-020-tr.html>)
- Install the transmission and all related components (if equipped). Refer to the OEM instructions
- If equipped with a wet flywheel housing, fill the flywheel housing with oil. Refer to OEM instructions
- If previously removed, attach any OEM attached components (mufflers, shift mechanisms, air filters, etc.) to the flywheel housing. Refer to the OEM instructions.



ck800wa

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



ck800wa

- Connect the batteries. Refer to the OEM service manual.
- Operate engine and check for leaks.